INTRODUCTION TO NETWORK DEVICE THEORY

1. Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

Network Devices in a Home Wi-Fi Setup:-

Device	Main Function	Typical Ports / Protocols
Switch	Connects multiple wired devices within a LAN.	Ethernet ports
Router	Connects the home network to the internet and routes data between networks.	WAN/LAN Ethernet ports, IP, NAT
Hub	Connects multiple devices and broadcasts incoming data to all ports.	Ethernet ports
Firewall	Protects the network by filtering incoming and outgoing traffic.	TCP 80 (HTTP), 443 (HTTPS), 53 (DNS), depending on allowed rules
Access Point	Connects wireless devices to the wired network through Wi-Fi.	Ethernet port, Wi-Fi (802.11), 2.4 GHz / 5 GHz bands

2. Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

Router:-

Ethernet/WAN Port: Connects to the internet.

LAN Port: Connects computers.

Power Port: Supplies power.

Antenna: Provides Wi-Fi signals.



2. Network Switch:-

Ethernet Ports: Connect network devices.

Console Port: Configures the switch.

Power Port: Supplies power.

3. Wireless Access Point (AP):-

Ethernet Port: Connects to the network.

Power Port: Supplies power.

Antenna: Provides wireless connectivity



3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

App: Instagram

1. Router: Connects my home Wi-Fi network to the internet so I can access Instagram.
2. Access Point: Connects my phone wirelessly to the local network through Wi-Fi.
3. Firewall: Helps protect the network by filtering unwanted or suspicious traffic.
4. Switch: Connects wired devices within the network and forwards data to the correct device.

4. Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Hint: Think about device capacity, coverage, and traffic handling.

Reason: The College Wi-Fi may be slow because the access points cannot handle too many connected students at the same time, causing congestion and disconnections.

Suggested Upgrade: Upgrade to modern, high-capacity Wi-Fi access points with better coverage and support for more devices.